

Use-Case Flow Specification

Use Case: Generate Optimized Collection Route

Preconditions

1. Smart bins have sent their current fill-level data to the system.
2. The sanitation supervisor is logged into the system.
3. The system has the current list of bins and their fill levels.

Postconditions

1. An optimized collection route is generated for the required bins.
2. The generated route is available to the truck driver.
3. The selected bins are included in the collection route.

Main Success Scenario

1. Sanitation Supervisor selects "**Generate Optimized Collection Route**".
2. System retrieves the latest bin fill-level data.
3. System filters the bins based on the **80% fill threshold**.
4. System identifies the bins that require collection.
5. System calculates an optimized collection route.
6. System displays the generated route to the Sanitation Supervisor.
7. System makes the route available to the Truck Driver.

Alternate Flow — Insufficient/Invalid Bin Data

1. System detects that the latest fill-level data for one or more bins is unavailable or invalid.
2. System excludes those bins from automatic route generation.
3. System displays a warning to the Sanitation Supervisor.
4. Sanitation Supervisor can manually tag a bin if it needs to be included.
5. System regenerates the optimized route with the manually tagged bin included.

